

1. Administrative & Study Identification

Full Study Title: A Randomised, Double-Blind, Placebo-Controlled, Parallel Group Study to Assess the Efficacy and Safety of Baxdrostat in Participants With Uncontrolled Hypertension on Two or More Medications Including Participants With Resistant Hypertension **Short Title/Acronym:** BaxHTN **Protocol Number:** D6970C00002 **NCT Number:** NCT06034743 **Sponsor Name:** AstraZeneca **Investigational Product:** Baxdrostat (1 mg or 2 mg) **Phase of Development:** Phase III **Medical Monitor:** AstraZeneca Clinical Operations Lead / Laura Munoz

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the difference in systolic blood pressure (SBP) change from baseline at Week 12 of treatment between participants receiving 1 mg or 2 mg baxdrostat tablets and participants receiving placebo tablets. **Secondary Objectives:** To evaluate various other assessments of blood pressure (including ambulatory BP), as well as the safety and tolerability of baxdrostat (monitoring for hyperkalemia, hyponatremia, and hypotension).

2.2 Background & Rationale To address the critical therapeutic gap in patients with uncontrolled or resistant hypertension who do not achieve target blood pressure despite adherence to multiple evidence-based therapies. Baxdrostat is a selective aldosterone synthase inhibitor (ASI) that blocks the production of aldosterone, a hormone that drives elevated blood pressure and increases cardiovascular and renal risk. This study assesses its efficacy as an add-on therapy.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Placebo-controlled **Blinding:** Double-blind (Quadruple Masking) with subsequent open-label periods **Randomization:** Parallel Group

3.2 Planned Enrollment & Duration Target \$N\$: 796 actual enrolled participants (initially estimated at ~720) **Number of Sites:** Multicenter (14 EU countries, UK, USA, Canada, and others) **Estimated Study Start:** 22-Nov-2023 (Actual) **Estimated Primary Completion:** 21-May-2025 (Actual)

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Male or female participants ≥ 18 years old. 2. Mean sitting systolic blood pressure on automated office blood pressure measurement ≥ 140 mmHg and < 170 mmHg at Screening. 3. **uHTN subpopulation:** Stable regimen of 2 antihypertensive medications from different therapeutic classes (at least one must be a diuretic), at maximum tolerated dose OR **rHTN subpopulation:** Stable regimen of ≥ 3 antihypertensive medications (at least one must be a diuretic), at maximum tolerated dose. 4. Estimated glomerular filtration rate (eGFR) ≥ 45 mL/min/1.73m² at Screening.

4.2 Exclusion Criteria 1. Mean sitting systolic blood pressure ≥ 170 mmHg or mean seated diastolic blood pressure ≥ 110 mmHg. 2. Serum sodium level < 135 mmol/L at Screening. 3. Known secondary causes of hypertension (e.g., renal artery stenosis, uncontrolled hyperthyroidism/hypothyroidism, pheochromocytoma, Cushing's syndrome, aortic coarctation) or severe heart failure (NYHA Class IV).

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Baxdrostat 1 mg, Baxdrostat 2 mg, or matching Placebo administered once daily (QD). **Route of Administration:** Oral (Tablet). **Duration of Treatment:** Up to 60 weeks of study duration.

5.2 Concomitant & Prohibited Therapy Permitted: Stable background regimen of 2 or ≥ 3 standard-of-care antihypertensive medications (including diuretics) depending on whether the patient has uncontrolled (uHTN) or resistant hypertension (rHTN). **Prohibited:** Initiation of new antihypertensive agents outside the trial protocol without medical necessity, or treatments known to induce secondary hypertension.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -14 to 0	Informed Consent, Medical History, Automated Office BP measurement, eGFR and Serum Sodium Labs
Baseline	Day 0	Randomization, First Dose, Background Medication Verification
Interim 1	Week 12	Primary Efficacy Assessment: Automated seated SBP measurement, Safety Labs (Potassium/Sodium monitoring)
End of Study	Week 60	Final Vital Signs (BP), Exit Interview, Investigational Product Accountability

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Difference in seated systolic blood pressure change from baseline at Week 12 of treatment compared to placebo. **Measurement Method:** Attended automated office blood pressure measurement.

7.2 Secondary & Exploratory Endpoints 1. Additional assessments of blood pressure (e.g., Diastolic blood pressure, 24-hour ambulatory blood pressure monitoring). 2. Safety and tolerability endpoints (incidence of adverse events, particularly focusing on hyperkalemia, hyponatremia, and hypotension).

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous monitoring via scheduled site visits and laboratory assessments, specifically tracking changes in serum electrolytes (potassium, sodium).

Serious Adverse Events (SAEs): Standard 24-hour reporting timeline for any event resulting in hospitalization, persistent disability, or life-threatening conditions. **Data Monitoring**

Committee (DMC): Utilized for ongoing safety oversight given the mechanism of action (aldosterone synthase inhibition).

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) population for the primary efficacy endpoint (SBP reduction); Safety Set for all AE/SAE evaluations. **Sample Size Justification:** Target $N = 720$ (actual enrolled $N=796$) adequately powered to detect a statistically significant and clinically meaningful placebo-adjusted reduction in systolic blood pressure at Week 12 (approx. 8.7 to 9.8 mmHg reductions anticipated based on dosage). **Handling of Missing Data:** Standard multiple imputation or Mixed Models for Repeated Measures (MMRM) methods for continuous blood pressure variables.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.

Monitoring Plan: Scheduled onsite and remote monitoring visits to ensure integrity of automated blood pressure readings and laboratory sampling. **Audit Trail:** eCRF locking and comprehensive data clarification forms tracking protocol adherence and background antihypertensive regimen stability.

11. Study Identifiers & Governance

Primary ID: D6970C00002 **Registry Number:** NCT06034743 **Sponsor/Lead Organization:** AstraZeneca **Study Title:** BaxHTN **Official Regulatory Title:** A Randomised, Double-Blind, Placebo-Controlled, Parallel Group Study to Assess the Efficacy and Safety of Baxdrostat in Participants With Uncontrolled Hypertension on Two or More Medications Including Participants With Resistant Hypertension

12. Clinical Framework (Hypertension Specific)

Primary Condition: Uncontrolled Hypertension (uHTN) and Resistant Hypertension (rHTN) **Diagnosis Threshold:** Mean seated SBP ≥ 140 mmHg and < 170 mmHg despite background therapies **Medical Methodology:** Addition of Aldosterone Synthase Inhibitor (ASI) to standard-of-care background regimens **Optimization Target:** Significant reduction in seated Systolic Blood Pressure by Week 12

13. Study Procedures & Methodology

Management Pathway: Continued administration of background standard-of-care antihypertensives alongside once-daily oral administration of investigational product or placebo. **Education Protocol (HCP):** Training on proper technique for automated office blood pressure measurements and identification of target electrolyte imbalances. **Education Protocol (Patient):** Strict adherence instructions regarding background antihypertensive maintenance and trial medication. **Quality Audit Mechanism:** Source data verification for blood pressure records and concomitant medication logs.

14. Observation & Follow-Up Schedule

Total Duration: Up to 60 weeks **Onsite Visit Cadence:** Screening, Baseline, Week 12 (Primary Endpoint), and extended follow-up up to Week 60. **Remote Monitoring Frequency:** Standard intervals per protocol for safety checks. **Checklist Review Interval:** Routine checks of concomitant medications and adverse events at every clinical visit.

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	Vivek Bhalla, MD (Stanford Lead) /						
Site PI	____	04-Apr-2026	Clinical Operations Lead	Laura			
Munoz	____	04-Apr-2026	Lead Statistician	[Name]	____		
	04-Apr-2026						